

Proof-Grounded Patient-Specific Clinical Explanations from Knowledge-Graph Reasoning

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Abstract—Clinical decision-support outputs can lack an auditable link between patient observations, encoded knowledge, conclusions, and recommendations. We present the CKG Clinical Explanation Engine, a downstream layer for a frozen, training-free clinical knowledge-graph reasoner that converts patient inference states and disease knowledge into typed facts, explicit rule-application traces, provenance-linked conclusions, and policy-licensed recommendations. The design separates measurement availability, representation completeness, and disease-specific activation; consequently, observed zero-activation evidence is not treated as missing and partial representation is distinct from unobserved evidence. Optional language generation is restricted to symbolically licensed content. Across five usable workbooks (6,720 patients; 20,160 patient–disease traces; 1,021,440 feature-evidence rows), IG-range validity and knowledge provenance were 100%, numerical cross-sheet fidelity was 100% (120,960/120,960), and exported logical/report trace completeness was 100% (20,160/20,160). Availability–representation consistency was 99.7028% (1,018,404/1,021,440); all 3,036 disagreements were confined to three systematic feature–cohort patterns. The corpus contained 86,783 observed zero-activation and 139,949 observed partially represented instances. A separate seeded 25-patient end-to-end audit completed without execution failure and passed all pre-specified trace, licensing, provenance, and state-consistency checks. These results establish structural and implementation auditability, not clinical correctness or utility.

Index Terms—clinical knowledge graph, symbolic reasoning, explainability, provenance, auditability, clinical decision support

I. INTRODUCTION

Clinical decision support increasingly combines statistical, knowledge-based, and language technologies, but a clinically readable explanation is not necessarily an auditable one. A useful explanation should expose how patient observations, encoded domain knowledge, intermediate conclusions, and permitted actions are connected. Post-hoc explanations can describe model behavior, while large language models (LLMs) can improve presentation, but neither property alone guarantees that every reported claim or recommendation is licensed by the underlying reasoning state [2], [3], [4], [5], [6].

This work introduces the *CKG Clinical Explanation Engine*, a downstream explanation layer for the training-free, ontology-grounded clinical knowledge-graph (CKG) reasoner developed by Das [1]. The upstream diagnostic computation is frozen: the explanation engine neither retrains nor recalibrates it and does not alter diagnostic scores or mappings.

Instead, it converts existing patient–disease inference outputs and disease knowledge into typed facts, rule-application traces with explicit premises, provenance-linked conclusions, and recommendations constrained by a closed policy.

The contribution is threefold. First, the engine separates measurement availability, representation completeness, and disease-specific activation, preventing a zero activation value from being interpreted as missing data. Second, it produces inspectable symbolic traces linking facts to conclusions and licensed actions. Third, we evaluate the frozen implementation at corpus scale and through an end-to-end engine audit. Optional language realization occurs only after symbolic licensing and cannot independently introduce diagnoses, measurements, tests, medications, or recommendations. Extended algorithms, audit tables, traces, and implementation details are provided in the supplementary material (<https://bit.ly/3VpWvby>).

II. RELATED WORK AND POSITIONING

Explainability in high-stakes settings is often discussed in terms of post-hoc accounts of model behavior [2], [3]. Such explanations can be useful, but they are different from an execution record showing which encoded facts and rules licensed a particular conclusion. Clinical LLM studies likewise demonstrate substantial capability while documenting limitations in factuality, calibration, and decision-making [4], [5], [6]. These observations motivate a constrained role for language generation rather than its use as the authority that determines clinical content.

Knowledge-graph-based clinical decision-support research has also used explicit semantic structure to improve transparency and expose relationships underlying recommendations [7]. More generally, provenance models such as W3C PROV provide a standardized vocabulary for representing how entities and activities contribute to derived information [8]. The present work addresses a narrower implementation problem: preserving an auditable patient-specific path from frozen upstream reasoning outputs through distinct availability, representation, and activation states to explicit rule applications and policy-licensed actions.

The present work is therefore positioned as a *downstream symbolic audit layer*. It does not propose another diagnostic classifier and does not compare predictive discrimination with outcome-fitted models. The diagnostic mapping is inherited

TABLE I
FROZEN UPSTREAM INTERFACE USED BY THE EXPLANATION ENGINE.

Input	Role in explanation layer
Patient value / mapping	Establishes the observed patient fact and its mapped ontology feature.
Clinical role	Preserves the disease-specific importance category encoded upstream.
$IG_D(p, f)$	Disease-specific feature activation; never used as a missingness flag.
Coverage a_{pf}	Describes completeness of encoded representation for an observed feature.
Disease score / evidence	Reported as immutable upstream outputs; not recomputed.
YAML provenance	Resolves mapped features and knowledge statements to packaged sources.
Policy entry	Enumerates the tests that may be recommended for eligible unresolved evidence.

unchanged from the upstream CKG reasoner of Das [1]. The contribution examined here is whether a patient-specific reasoning state can be transformed into an inspectable explanation in which observations, representation status, disease-specific activation, conclusions, provenance, and permitted recommendations remain distinguishable. Here, language realization is deliberately downstream of the symbolic authorization boundary: the evaluated deterministic path fixes the clinical content before any optional natural-language realization.

III. METHOD

A. Relationship to the Frozen CKG Reasoner

For patient p , disease D , and feature f , the upstream system of Das [1] exports patient values, feature mappings, clinical roles, disease-level scores, evidence coverage, and a disease-specific information-gate value $IG_D(p, f) \in [0, 1]$. The explanation layer treats these quantities as immutable facts. Disease-level scores are therefore evidence to be explained, not quantities recomputed by the explanation engine.

The processing chain is

CKG output + knowledge/policy $\rightarrow$ typed facts $\rightarrow$ rule traces
 $\rightarrow$ licensed claims/actions $\rightarrow$ report.

The boundary between the two systems is intentional. The upstream reasoner owns disease scoring and evidence activation; the explanation engine owns only interpretation and trace construction. Table I summarizes the interface used by the explanation layer. A change to the explanation renderer therefore cannot feed back into the diagnostic score.

B. Orthogonal Evidence State

The original implementation contains information about whether a measurement exists, how completely it is represented by the encoded components, and whether it activates disease-specific evidence. These concepts are kept separate.

Let $m_{pf} \in \{0, 1\}$ denote measurement availability and $a_{pf} \in [0, 1]$ representation coverage. Then

$$\text{Observed}(p, f) \Leftrightarrow m_{pf} = 1, \quad (1)$$

$$\text{Unobserved}(p, f) \Leftrightarrow m_{pf} = 0, \quad (2)$$

$$\text{Unrepresented}(p, f) \Leftrightarrow m_{pf} = 1 \wedge a_{pf} = 0, \quad (3)$$

$$\text{Partial}(p, f) \Leftrightarrow m_{pf} = 1 \wedge 0 < a_{pf} < 1, \quad (4)$$

$$\text{Full}(p, f) \Leftrightarrow m_{pf} = 1 \wedge a_{pf} = 1. \quad (5)$$

The activation $IG_D(p, f)$ is disease-specific and is not an availability flag. In particular, $IG_D(p, f) = 0 \not\Rightarrow \text{Unobserved}(p, f)$. Thus an observed measurement may contribute zero disease-specific activation, and an observed feature may simultaneously be partially represented. This formulation removes the ambiguity created by treating “observed,” “partial,” and “unobserved” as competing states.

C. Rule-Application Proof Traces

The engine forms a fact set $\Gamma_{p,D}$ containing patient values, mappings, roles, availability, coverage, activation, and provenance. Rules consume explicit premises and emit a conclusion together with a rule identifier and reason. Representative rules are

$$\text{Observed}(p, f) \wedge \text{Role}_D(f, r) \wedge IG_D(p, f) \geq \tau \Rightarrow \text{EncodedSupport}(p, D, f), \quad (6)$$

$$\text{Observed}(p, f) \wedge 0 < a_{pf} < 1 \Rightarrow \text{PartiallyRepresented}(p, f), \quad (7)$$

$$\text{Unobserved}(p, f) \wedge \text{CriticalRole}_D(f) \Rightarrow \text{CriticalIncomplete}(p, D, f), \quad (8)$$

$$\text{CriticalIncomplete}(p, D, f) \wedge \text{LicensedTest}_D(f, t) \Rightarrow \text{Recommend}(p, t). \quad (9)$$

Here “proof” denotes an inspectable rule-application certificate: each exported step records its premises, rule identifier, conclusion, and reason/provenance. It does *not* denote independent proof-assistant verification of logical soundness or clinical truth. Recommendations are emitted only when the symbolic state and the explicit recommendation policy jointly license them. Low coverage, partial representation, or an LLM cannot independently create a test recommendation.

A deterministic renderer is the evaluated reporting path. An optional LLM may realize already licensed content in natural language, subject to checks that numerical values originate from facts and actions belong to the policy; medication and dose generation are disabled. The LLM path is not evaluated here.

This creates a closed recommendation boundary. For example, an observed feature with $IG_D(p, f) = 0$ can be described as observed but non-activating; it cannot be relabelled “missing.” Likewise, partial representation can be surfaced as an uncertainty in encoded coverage, but it cannot itself authorize a test. A test action requires both an unresolved genuinely unobserved feature and a matching policy entry, as in (9). Consequently, the explanatory state may expose uncertainty without automatically converting that uncertainty into an intervention.

Each rule application is exported with four audit-relevant fields: premises, rule identifier, conclusion, and reason/provenance. Trace completeness in this paper means that

these required fields are present for the exported rule application. It does not imply that an external theorem prover has certified the rule base. This operational definition is used consistently in the evaluation below.

D. Trace Construction and Audit Contract

For each patient–disease pair, the explanation engine constructs the trace from already exported feature-level evidence rather than reconstructing the upstream diagnostic calculation. A patient value is first associated with its mapped feature and disease-specific role. Availability is then read independently from representation coverage and activation. This ordering is important: an *IG* value is interpreted only after the system knows whether the underlying measurement was observed, and coverage is interpreted as a property of representation rather than acquisition. The resulting typed facts form the premises consumed by the explanation rules.

A trace step is considered structurally complete only when the exported artifact contains the premises consumed by the rule, a rule identifier, the resulting conclusion, and a human-readable reason or provenance reference. The deterministic report is generated from these trace-level conclusions rather than directly from raw workbook cells. Thus, the report can be audited backwards from a sentence or action to a conclusion, from the conclusion to a rule application, and from the rule application to patient and knowledge facts. This is the operational sense in which the explanation is “proof-grounded.”

The recommendation policy is deliberately narrower than the descriptive trace. The trace may describe observed support, observed non-activation, partial representation, or unresolved evidence, but only the final category can participate in a confirmatory-test rule, and then only when a disease/feature/test mapping exists in the packaged policy. This separates *describing uncertainty* from *authorizing an action*. It also provides a concrete verification target: every emitted recommendation must resolve to a policy identifier and every cited knowledge feature must resolve to packaged YAML provenance.

IV. EVALUATION

The evaluation used the frozen package and a separate reproducibility workflow without modifying upstream diagnostic inference. Five non-empty CKG output workbooks were usable; an additional zero-byte influenza workbook was excluded before analysis. Together they contained 6,720 patients, three disease traces per patient (20,160 traces), and 1,021,440 feature-evidence rows.

The full-corpus audit checked: (i) availability/representation consistency; (ii) $IG \in [0, 1]$; (iii) coverage/component consistency; (iv) resolution of knowledge provenance to packaged YAML; (v) numerical agreement across exported sheets; and (vi) completeness of exported logical and deterministic-report traces. These are structural implementation invariants, not tests of clinical validity.

A second, seeded end-to-end audit (seed 20260926) invoked the current patient reasoner for five patients sampled independently within each usable workbook ($n = 25$); workbook-specific seeded sampling makes the selected identifiers reproducible across reruns. It checked successful execution, required proof-step fields, recommendation identifiers against the closed policy, YAML-source resolution, and consistency between availability and representation states. The audit retained exceptions rather than silently dropping failed cases. Full workbook-level results, patient identifiers, traces, runtime data, and failure files are placed in the supplementary material (<https://bit.ly/3VpWvby>).

The two audit levels answer different questions. The corpus audit tests exported structures at scale, whereas the end-to-end audit executes the current explanation path and checks the resulting report artifacts. Keeping these levels separate avoids treating successful parsing of historical exports as equivalent to successful execution of the current engine. No outcome label is used by the explanation layer to create or validate a recommendation.

A. Audit Independence, Regression Checks, and Statistical Interpretation

The reproducibility workflow is implemented separately from the patient-report engine and reads the frozen workbooks and packaged YAML artifacts directly. This separation reduces dependence on the report renderer when checking workbook invariants, numerical cross-sheet agreement, and provenance resolution. It is nevertheless *not* an independent formal verifier: the workflow and engine necessarily share the documented semantics of fields such as *Observed*, *coverage*, and *policy identifiers*. Accordingly, the reported pass rates are described as structural consistency and implementation checks, not independent certification of logical or clinical correctness.

As an additional software-regression check, the supplied frozen package was executed against its focused patient-report, evidence-state, and FOL/HOL reasoner tests; all six selected tests passed. These tests exercise report construction and the evidence-state logic but are not counted as patient-level validation cases. Deliberate mutation testing of proof objects was not part of the frozen experiment and is therefore not claimed here; adversarial mutation of premises, rule identifiers, provenance, and policy identifiers remains a useful future validation of validator sensitivity.

The corpus percentages are descriptive proportions over exported rows. Feature rows are nested within patients and workbooks and therefore should not be treated as independent Bernoulli trials. For this reason, no binomial confidence interval is attached to the 99.7028% consistency rate in the revised analysis. The seeded end-to-end audit likewise reports exact pass counts rather than estimating a population-level clinical error rate.

We additionally retained two naturally occurring stress conditions rather than filtering them: observed features with $IG = 0$ and observed features with $0 < a_{pf} < 1$. These cases directly test the state semantics because collapsing activation,

TABLE II
FROZEN-SYSTEM STRUCTURAL AUDIT.

Check	Passed / Total	Rate
Availability–representation consistency	1,018,404 / 1,021,440	99.7028%
IG range validity	1,021,440 / 1,021,440	100%
Knowledge provenance resolution	1,021,440 / 1,021,440	100%
Numerical cross-sheet fidelity	120,960 / 120,960	100%
Exported logical trace completeness	20,160 / 20,160	100%
Deterministic-report trace completeness	20,160 / 20,160	100%

representation, and availability would misclassify them. The failure table was also retained in full so that systematic disagreements could be localized by workbook, disease, and feature instead of being hidden behind an aggregate pass rate.

V. RESULTS

Table II summarizes the corpus audit. Importantly, the 3,036 state disagreements were not heterogeneous failures. They were confined to three systematic feature–cohort combinations: 1,018 `VOM_manifestation` records in Dengue traces and 1,018 `VOM_progression` records in Malaria traces from the same 1,018-patient workbook, plus 1,000 `MAL_manifestation` records in Dengue traces from a 1,000-patient workbook. In every case the feature-evidence export marked the measurement as observed with $IG = 0$, whereas the confidence audit recorded zero representational coverage and `status=missing`. This is therefore an availability–representation disagreement between exported structures, not evidence that the current explanation engine converts the measurement to unobserved. The current engine determines availability from the feature-evidence `Observed` field; recommendations additionally require a genuinely unobserved policy-listed confirmatory feature. The disagreement is retained and reported rather than overwritten.

This localization changes the interpretation of the aggregate 99.7028% rate. The failures do not indicate that approximately 0.3% of arbitrary features are unpredictably corrupted. Instead, they identify three repeated schema-level combinations in which two exported structures encode different dimensions using superficially similar missingness terminology. Each pattern repeats once per corresponding patient record. Because the explanation engine reads availability from `Feature_Evidence.Observed`, a zero-coverage `Confidence_Audit` record does not by itself satisfy the `Unobserved` premise of rule (8). The audit therefore preserves the discrepancy as an export inconsistency while preventing it from being silently reinterpreted as absent measurement. Correcting the upstream export terminology remains preferable, but doing so is outside the frozen-system evaluation reported here.

The downstream impact is bounded by the current engine’s rule premises. The three disagreement patterns cannot, by themselves, satisfy `Unobserved` because their feature-evidence records are explicitly observed. They therefore cannot independently trigger the unresolved-critical-evidence rule or a confirmatory-test recommendation. This

TABLE III
ILLUSTRATIVE STATE DISTINCTIONS RETAINED FOR PATIENT 152.

Evidence	State	Explanation consequence
Hemoconcentration	Observed, $IG = 1.000$	Retained as activated high-role evidence.
Mild thrombocytopenia	Observed, $IG \approx 0.953$	Retained as activated high-role evidence.
Leukopenia / neutropenia	Observed, $IG = 0$	Remains observed; not converted to missing.
Partially encoded high-role evidence	Observed, partial	Coverage limitation exposed without creating a test action.
NS1 / PCR evidence	Unobserved	Eligible for confirmatory-test licensing only through policy.

is a code-path conclusion about the current engine, not evidence that the legacy export is harmless for every external consumer: software that instead interprets `Confidence_Audit.status=missing` as acquisition missingness could reach a different state. The disagreement is therefore retained as a source-data/schema defect and not normalized away.

The corpus also contained 86,783 observed feature instances with $IG = 0$ and 139,949 observed instances with partial representation. These stress cases demonstrate why availability, representation, and activation must not be collapsed into a single missingness state.

All 25 end-to-end cases executed without exception and passed the pre-specified proof-completeness, recommendation-licensing, source-resolution, and availability/representation checks. Mean execution time varied across the five workbooks from approximately 14.8 to 52.0 s per audited patient in the recorded experimental environment; these measurements characterize the audited implementation rather than a general deployment benchmark.

For an illustrative patient (ID 152), the frozen upstream Dengue trace reported evidence probability 0.1179, Hallmark/Major cosine similarity 0.7062, relative magnitude 0.6737, disease score 0.4993, and evidence coverage 0.3715. Hemoconcentration and mild thrombocytopenia were observed high-role evidence with $IG = 1.000$ and approximately 0.953, respectively, whereas leukopenia/neutropenia remained observed despite $IG = 0$. Other high-role evidence was partially represented, while NS1/PCR evidence was genuinely unobserved. The rule trace therefore preserves these distinctions before the closed policy licenses confirmatory testing. The complete machine-readable trace is supplied as supplementary material.

VI. DISCUSSION

The evaluation supports three implementation-level conclusions. First, measurement availability cannot be inferred from disease-specific activation: tens of thousands of observed instances legitimately had zero activation. Second, partial representation is a property of how an observed feature is encoded, not a synonym for missing measurement. Third, provenance and numerical fidelity can be checked mechanically across the exported reasoning artifacts.

The systematic 3,036 disagreements are also informative. The earlier binary audit conflated an observed measurement with a representation record labelled `missing`. The revised formulation exposes the disagreement instead of forcing the two dimensions into one state. Because the current engine obtains availability from `Feature_Evidence.Observed`, these rows do not by themselves become unobserved evidence; nevertheless, the legacy export disagreement should be corrected at its source and remains visible in the supplementary failure table.

These results also clarify what the explanation layer contributes beyond a formatted report. A conventional renderer can reproduce values, but it need not preserve the distinction between a missing observation, an incompletely represented observation, and a present observation that contributes no disease-specific activation. Here those distinctions are explicit predicates and are therefore available to both the deterministic report and the reproducibility checks. This is particularly important in the 86,783 zero-activation and 139,949 partial-representation instances, where a single generic “missing/weak evidence” label would discard materially different computational states.

The recommendation boundary provides a second form of auditability. Recommendations are not inferred from narrative plausibility; they are consequences of a symbolic unresolved-evidence state plus a closed policy lookup. This makes the provenance of an action inspectable and prevents the optional language layer from expanding the action space. The design is intentionally conservative: a clinically sensible action absent from the policy is not silently invented, but instead requires an explicit knowledge/policy update outside the patient-level explanation run.

The corpus and end-to-end audits should also be interpreted together. Corpus-scale checks provide breadth over more than one million feature-evidence rows, while the seeded engine audit verifies that the current executable pathway can construct complete trace and report artifacts for sampled patients. The former is stronger for detecting systematic export inconsistencies; the latter is stronger for detecting execution, policy-resolution, or report-construction failures. Neither substitutes for clinical validation.

The claims are deliberately bounded. Passing a rule-trace or provenance check does not establish that a medical rule, threshold, knowledge entry, or recommendation policy is clinically correct. The study is cross-sectional and does not evaluate prognosis, patient outcomes, clinician decision quality, or prospective clinical utility. The 25-patient engine audit is smaller than the corpus-level exported-structure audit and should not be interpreted as exhaustive end-to-end clinical validation. Likewise, no claim is made that the constrained architecture outperforms unrestricted LLM reporting; such a comparison addresses a separate empirical question. Clinical review, prospective validation, and broader deployment testing remain necessary.

A. Validity Boundaries

There are three distinct validity layers. *Structural validity* asks whether exported values, provenance, trace fields, and policy identifiers are internally consistent; this is the principal target of the present experiments. *Semantic validity* asks whether the encoded medical roles, thresholds, and rules faithfully represent the intended clinical knowledge. *Clinical validity* asks whether using the resulting explanation improves decisions or patient outcomes. Passing the first layer does not establish the latter two. This distinction is important because a perfectly traceable system can still expose an incorrect encoded rule; traceability makes that error inspectable rather than clinically correct.

The current evaluation also inherits the scope of the frozen artifacts. It is cross-sectional, uses the available disease knowledge and policy files, and evaluates the deterministic explanation path. The optional LLM realization layer is intentionally excluded from the reported pass rates. The end-to-end sample is reproducibly seeded but smaller than the corpus audit; its role is to exercise the executable pathway, not to estimate a clinical error rate. These boundaries motivate future clinician review of rule/policy content and prospective evaluation without changing the implementation-level conclusions tested here.

B. Reproducibility Details

The audit notebook fixes random seed 20260926 and records the selected patient identifiers, per-case status, proof-step count, recommendation count, and execution time. Exceptions are written to dedicated failure files rather than removed from denominators. The corpus audit likewise writes every failed invariant to a row-level failure table containing workbook, patient/disease/feature context, test name, and diagnostic detail. This design permits aggregate numbers in the paper to be traced back to the underlying exported record. The zero-byte influenza workbook is excluded before analysis and is not included in any denominator.

C. Reproducibility Materials

To keep the conference paper within the conference page limit, the Reproducibility Materials contain the workbook-level structural-check tables, the complete 3,036-row failure localization, sampled patient identifiers, machine-readable proof traces, runtime records, YAML knowledge/policy files, and experiment scripts/notebooks. The main paper reports only quantities that are directly supported by those frozen artifacts. The Reproducibility Materials are intended to permit inspection of individual failures and trace steps without changing the scope of the claims made here (<https://bit.ly/3VpWvby>).

VII. CONCLUSION

We presented a downstream explanation engine that leaves a training-free CKG diagnostic reasoner unchanged while converting its patient-specific state into typed facts, explicit rule-application traces, provenance-linked conclusions, and policy-licensed recommendations. Across 6,720 patients and 20,160 patient–disease traces, the implementation showed complete

IG-range, provenance, numerical-fidelity, and exported-trace checks, while a 99.7028% availability–representation consistency audit localized all remaining disagreements to three systematic export patterns. Separating availability, representation completeness, and disease-specific activation prevents observed zero-activation or partial evidence from being misreported as absent. These findings establish structural auditability of the implemented explanation pathway; they do not establish clinical correctness or utility.

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